

Liang Gao

Plainsboro, NJ | gaoliang715@gmail.com | 504-312-8559 | linkedin.com/in/lianggao9 | hslgao4.github.io

Education

- George Washington University**, MS in Data Science | GPA: 3.9/4.0 Jan 2023 – Dec 2024
- **Coursework:** Time Series Modeling & Forecasting, Machine Learning, Data Visualization, Deep Learning, NLP
- Tulane University**, MS in Finance | GPA: 3.7/4.0 Aug 2021 – Dec 2022
- **Coursework:** Financial Modeling, Fixed Income Analytics, Accounting Analytics
- Xi'an Jiaotong-Liverpool University | University of Liverpool** Sep 2017 – Jun 2021
- Bachelor of Arts – Business Management (2+2 Program)
- **Coursework:** Quantitative Methods, Financial Management, International Finance

Work Experience

- Data Science Contract** Mar 2025 – Present
- Siemens Healthineers, NJ
- Automated data preprocessing pipelines in Python for 10k+ medical data, replacing manual steps with optimized scripts and reducing recurring workload by 5+ hours per week.
 - Engineered high-fidelity datasets for AI model training; accelerated the development cycle by 20% through robust ETL workflows and automated data validation protocols.
 - Executed systematic model evaluation and analysis on 40+ datasets; conducted t-tests and visualizations to quantify model performance and extracted actionable insights that prioritized downstream retraining strategies.
 - Developed automated QA reporting tools using Python to visualize 3D datasets; generated HTML-based diagnostic reports to streamline manual verification and support data-driven decision-making.
- Graduate Teaching Assistant** Sep 2024 – Dec 2024
- George Washington University, DC
- Explained statistical and machine learning concepts to audiences with diverse technical backgrounds, strengthening the ability to communicate complex analyses to non-technical stakeholders.
 - Automated grading and evaluation workflows using Python, improving assessment consistency and reducing manual processing time by 40%.

Selected Projects

- Time Series Models Benchmarking | Python** Aug 2024 – Dec 2024
- Benchmarked ARIMA, LSTM, and Transformer architectures for enterprise-scale forecasting, identifying optimal models to support scenario planning and operational decision-making.
 - Conducted exploratory time-series analysis, including ADF stationarity tests and ACF/PACF mapping, to determine optimal model parameters and data transformations.
 - Deployed models on AWS EC2, utilizing scalable data workflows and evaluating performance via backtesting and error metrics (e.g., RMSE, MAE).
- Deep Learning-Based Classification & Segmentation | Python, PyTorch** Jan 2024 - May 2024
- Developed and optimized a CNN-based (U-Net) architecture for high-precision biomedical image classification and segmentation, a Dice similarity coefficient of 0.93.
 - Optimized model performance via hyperparameter tuning and data augmentation, reducing training time and improving convergence stability without sacrificing precision.
- AI Book Recommendation System | Python, PyTorch** Jan 2024 - May 2024
- Processed and analyzed text data from 50k+ book reviews using NLP techniques (NLTK, Spacy).
 - Developed a book recommendation system in Python utilizing Transformer architecture, specifically leveraging pre-trained BERT models with Pytorch. Designed an interface using Streamlit and deployed it on AWS.

Technical Skills

Programming & Tools: Python, SQL(PostgreSQL, MySQL), Git

Machine Learning: Regression, XGBoost, Time Series Modeling, Computer Vision, NLP

Statistics: Hypothesis Testing, A/B Testing

Data & Visualization: Pandas, NumPy, Scikit-learn, Matplotlib, Power BI(Microsoft Press)